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Recessed vertical interconnects for device miniaturization

Abstract

The present disclosure relates to an electronic assembly including a package substrate with a first surface and an opposing second surface; a first interconnect disposed in the package substrate and extending between the first and the second surfaces; and a second interconnect disposed in the package substrate and extending between the first and the second surfaces; wherein the first interconnect comprises a first recessed side wall and the second interconnect is arranged adjacent the first recessed side wall.

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Background/Summary

BACKGROUND

(1) Current solutions to address package and/or platform miniaturization through vertical interconnects scaling include reduced plated through hole (PTH) drill size, e.g., drill bit diameter reduction from 10 mils to 6 mils, and/or improved mechanical registration accuracy between vertical interconnects and contact pads from 4 mils to 2.5 mils through improved imaging process.

(2) Large PTH interconnects pitch geometry (i.e., distance between centers of two adjacent PTH interconnects) impacts the package and/or printed circuit board routing density and flexibility, thus inhibiting device miniaturization.

(3) The disadvantages of the above-mentioned solutions include manufacturing costs trade-off, i.e., increased frequency of mechanical drill bit replacement with reduced bit diameter and increased assembly yield losses due to more stringent screening of parts meeting the desired mechanical registration specifications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings, like reference characters generally refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating the principles of the present disclosure. The dimensions of the various features or elements may be arbitrarily expanded or reduced for clarity. In the following description, various aspects of the present disclosure are described with reference to the following drawings, in which:

(2) FIG. **1**A shows a perspective view of an electronic assembly including recessed vertical interconnects for improved electrical performance and device miniaturization according to an aspect of the present disclosure;
(3) FIG. **1**B shows a top view of the electronic assembly according to the aspect as shown in FIG. **1**A;
(4) FIG. **1**C shows a cross-sectional view of the electronic assembly according to the aspect as shown in FIG. **1**A;
(5) FIG. **1**D shows PTH vertical interconnects of (i) a conventional design, and presently disclosed recessed design with (ii) a Recess Ø=10 mils and (iii) a Recess Ø=12 mils;
(6) FIG. **1**E shows the electrical performance comparison in insertion losses for the PTH vertical interconnects designs of FIG. **1**D;
(7) FIG. **2**A shows a perspective view of recessed vertical interconnects in an electronic assembly for improved electromagnetic field coupling according to another aspect of the present disclosure;
(8) FIG. **2**B shows a top view of the electronic assembly according to the aspect as shown in FIG. **2**A;
(9) FIG. **2**C shows a cross-sectional view of the electronic assembly according to the aspect as shown in FIG. **2**A;
(10) FIGS. **3**A through **3**B show views directed to an exemplary simplified process flow for manufacturing the recessed vertical interconnects according to an aspect that is generally similar to that shown in FIG. **1**A of the present disclosure;
(11) FIGS. **4**A through **4**G show views directed to an exemplary simplified process flow for manufacturing the recessed vertical interconnects according to an aspect that is generally similar to that shown in FIG. **2**A of the present disclosure;
(12) FIG. **5** shows an illustration of a computing device that includes an electronic assembly according to a further aspect of the present disclosure; and
(13) FIG. **6** shows a flow chart illustrating a method for forming an electronic assembly according to an aspect of the present disclosure.

DETAILED DESCRIPTION

(14) The following detailed description refers to the accompanying drawings that show, by way of illustration, specific details and aspects in which the present disclosure may be practiced. These aspects are described in sufficient detail to enable those skilled in the art to practice the present disclosure. Various aspects are provided for devices, and various aspects are provided for methods. It will be understood that the basic properties of the devices also hold for the methods and vice versa. Other aspects may be utilized and structural, and logical changes may be made without departing from the scope of the present disclosure. The various aspects are not necessarily mutually exclusive, as some aspects may be combined with one or more other aspects to form new aspects.
(15) This present disclosure addresses the package and/or printed circuit board real-estate miniaturization due to the limitation of existing vertical interconnects, e.g., plated through hole (PTH) interconnects pitch scaling.
(16) The present disclosure generally relates to an electronic assembly. The electronic assembly may include a package substrate with a first surface and an opposing second surface. The electronic assembly may include a first interconnect in the package substrate and the first interconnect may extend between the first and the second surfaces. The electronic assembly may include a second interconnect in the package substrate and the second interconnect may extend between the first and the second surfaces. The first interconnect may include a first recessed side wall and the second interconnect may be arranged adjacent the first recessed side wall.
(17) In various aspects, the first interconnect, the second interconnect, or both, may extend between the first (e.g., top) and the second (e.g., bottom) surfaces of the package substrate to form vertical interconnects.
(18) The first interconnect may include a side wall that may extend along a length (partially or

fully) thereof. The first recessed side wall may include a receding portion that cuts inwards. The receding portion may extend along a length (partially or fully) of the side wall. In various aspects, the first interconnect may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible.

(19) In various aspects, the electronic assembly may further include a first contact pad including a first recessed side coupled to a first terminal of the first interconnect. The electronic assembly may also include a subsequent first contact pad including a subsequent first recessed side coupled to an opposing second terminal of the first interconnect. The first and the subsequent first recessed sides may be aligned with the first recessed side wall.

(20) The first recessed side of the first contact pad may include a receding portion that cuts inwards. In various aspects, the first contact pad may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the first recessed side wall and the first recessed side may both have the same shape, e.g., a crescent shape. The same discussion may also apply to the subsequent first recessed side of the subsequent first contact pad.

(21) In various aspects, the second interconnect may be at least partially encircled by the first recessed side wall.

(22) In various aspects, the second interconnect may further include a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces.

(23) In various aspects, the second interconnect may include a side wall that may extend along a length (partially or fully) thereof. The second recessed side wall may include a receding portion that cuts inwards. The receding portion may extend along a length (partially or fully) of the side wall. In various aspects, the second interconnect may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. In other aspects, the second interconnect may not include a recessed side wall.

(24) In various aspects, the first and the second recessed sides may face each other.

(25) In various aspects, the electronic assembly may further include a second contact pad including a second recessed side coupled to a first terminal of the second interconnect. The electronic assembly may include a subsequent second contact pad including a subsequent second recessed side coupled to an opposing second terminal of the second interconnect. The second and the subsequent second recessed sides may be aligned with the second recessed side wall.

(26) The second recessed side of the second contact pad may include a receding portion that cuts inwards. In various aspects, the second contact pad may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the second recessed side wall and the second recessed side may both have the same shape, e.g., a crescent shape. The same discussion may also apply to the subsequent second recessed side of the subsequent second contact pad. In other aspects, the second contact pad, the subsequent second contact pad, or both, may not include a recessed side.

(27) In various aspects, the electronic assembly may further include a plurality of first metal traces or planes coupled to the first and the second contact pads, and further include a plurality of second metal traces or planes coupled to the subsequent first and the subsequent second contact pads.

(28) In various aspects, the electronic assembly may further include one or more devices coupled to the first and the second interconnects at the first surface.

(29) In various aspects, the one or more devices may include one or more semiconductor dies and/or packages, e.g., a central processing unit (CPU), a graphic processing unit (GPU), a memory controller, a field programmable gate array (FPGA), a neural network accelerator, a communication device e.g., RFIC or a platform controller hub or chipset.

(30) In various aspects, the first and second interconnects may be configured to facilitate a differential pair electrical signal.

(31) In various aspects, the first interconnect may be configured to facilitate a reference voltage and

the second interconnect may be configured to facilitate a single-ended electrical signal.
(32) The present disclosure also generally relates to a computing device. The computing device may include a communication chip and an electronic assembly coupled to the communication chip. The electronic assembly may include a package substrate with a first surface and an opposing second surface. The electronic assembly may include a first interconnect in the package substrate and the first interconnect may extend between the first and the second surfaces. The electronic assembly may include a second interconnect in the package substrate and the second interconnect may extend between the first and the second surfaces. The first interconnect may include a first recessed side wall and the second interconnect may be arranged adjacent the first recessed side wall.
(33) The present disclosure further generally relates to a method. The method may include providing a package substrate with a first surface and an opposing second surface. The method may include arranging a first interconnect in the package substrate, the first interconnect extending between the first and the second surfaces, wherein the first interconnect may include a first recessed side wall. The method may include arranging a second interconnect adjacent the first recessed side wall in the package substrate, the second interconnect extending between the first and the second surfaces.
(34) The technical advantages of the present disclosure may include but are not limited to: Device (package and/or printed circuit board) miniaturization through reduced PTH vertical interconnects pitch geometry; >20% package and/or board real-estate savings and/or z-height reduction (e.g., through reduced signal redistribution layers) achievable through improved routing density. Improved electrical (signal and/or power integrity) performance through tighter signal PTH to ground PTH coupling, i.e., improved signal current return path (for single-ended bus e.g., DDR memory interface) and tighter differential signal PTHs coupling, e.g., a universal serial bus (USB) interface, a peripheral component interconnect express (PCIe) interface or a high speed ethernet interface. Tighter power (Vcc) to ground (Vss) interconnect coupling reduces AC loop inductance for improved power supply delivery or power integrity.
(35) To more readily understand and put into practical effect the present disclosure, particular aspects will now be described by way of examples and not limitations, and with reference to the drawings. For the sake of brevity, duplicate descriptions of features and properties may be omitted.
(36) FIG. **1**A shows a perspective view of recessed vertical interconnects **108**, **112** in an electronic assembly **100** for improved electrical performance and device miniaturization according to an aspect of the present disclosure. FIG. **1**B shows a top view of the electronic assembly **100** according to the aspect as shown in FIG. **1**A. FIG. **1**C shows a cross-sectional view of the electronic assembly **100** according to the aspect as shown in FIG. **1**A.
(37) In various aspects, the electronic assembly **100** may include a package substrate **102** having a first surface **104** and an opposing second surface **106**. The first surface **104** may be a top surface of the package substrate **102**. The second surface **106** may be a bottom surface of the package substrate **102**. The first and second surfaces **104**, **106** may be substantially horizontal when viewed from a cross-sectional viewpoint. The package substrate **102** may further include a plurality of metal planes **138** formed by metal layers embedded therein. In one aspect, the package substrate **102** may include two or more metal layers spaced apart by a dielectric layer. In one aspect, the package substrate **102** is a printed circuit board.
(38) In various aspects, the package substrate **102** may include a first interconnect **108**. The first interconnect **108** may include a first side wall **110** having a length that may extend between the first and second surfaces **104**, **106** of the package substrate **102**. In one aspect, the first interconnect **108** may extend vertically across the thickness of the package substrate **102**. In other words, the first interconnect **108** may be perpendicular to the first and second surfaces **104**, **106** to form a first vertical interconnect **108**.
(39) In various aspects, the first side wall **110** may include a recessed portion that may extend

between the first and second surfaces **104**, **106** of the package substrate **102** to form a first recessed side wall **110**. In the aspect shown in FIG. **1**A and FIG. **1**B, the first recessed side wall **110** of the first interconnect **108** may have a crescent-shaped cross section.
(40) In various aspects, the package substrate **102** may include a second interconnect **112**. The second interconnect **112** may include a second side wall **114** having a length that may extend between the first and second surfaces **104**, **106** of the package substrate **102**. In one aspect, the second interconnect **112** may extend vertically across the thickness of the package substrate **102**. In other words, the second interconnect **112** may be perpendicular to the first and second surfaces **104**, **106** to form a second vertical interconnect **112**.
(41) In various aspects, the second side wall **114** may include a recessed portion that may extend between the first and second surfaces **104**, **106** of the package substrate **102** to form a second recessed side wall **114**. In the aspect shown in FIG. **1**A and FIG. **1**B, the second recessed side wall **114** of the second interconnect **112** may have a crescent-shaped cross section.
(42) In one aspect, the second recessed side wall **114** may extend in parallel with the first recessed side wall **110** between the first and second surfaces **104**, **106**. In an aspect, the first recessed side wall **110** may be configured to face the second recessed side wall **114**. In other words, the first interconnect **108** may be arranged adjacent or close to, but not contacting, the second interconnect **112**. The arrangement may resemble a mirror image of the two interconnects **108**, **112**. The first and the second interconnects **108**, **112** may be separated or kept apart by a recess opening **136** formed in the package substrate **102**. The distance between the centers of the first and the second interconnects **108**, **112** may be defined as a pitch.
(43) In various aspects, the electronic assembly **100** may further include a first contact pad **116** arranged on the first surface **104** of the package substrate **102**. The first contact pad **116** may include a first recessed side **118** coupled to a first terminal **140** of the first interconnect **108** at the first surface **104**.
(44) The first recessed side **118** of the first contact pad **116** may include a receding portion that cuts inwards. In various aspects, the first contact pad **116** may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the first recessed side wall **110** and the first recessed side **118** may both have the same shape, e.g., a crescent shape as shown in FIG. **1**A and FIG. **1**B.
(45) In various aspects, the electronic assembly **100** may further include a subsequent first contact pad **120** arranged on the second surface **106** of the package substrate **102**. The subsequent first contact pad **120** may include a subsequent first recessed side **122** coupled to an opposing second terminal **142** of the first interconnect **108** at the second surface **106**.
(46) The subsequent first recessed side **122** of the subsequent first contact pad **120** may include a receding portion that cuts inwards. In various aspects, the subsequent first contact pad **120** may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the first recessed side wall **110** and the subsequent first recessed side **122** may both have the same shape, e.g., a crescent shape as shown in FIG. **1**A and FIG. **1**B.
(47) In one aspect, the first and the subsequent first recessed sides **118**, **122** may be aligned with the first recessed side wall **110** of the first interconnect **108**. In other words, the respective receding portion of the first recessed side **118** and the first recessed side wall **110** may coincide at the first surface **104** and the respective receding portion of the subsequent first recessed side **122** and the first recessed side wall **110** may coincide at the second surface **106**.
(48) In various aspects, the electronic assembly **100** may further include a second contact pad **124** arranged on the first surface **104** of the package substrate **102**. The second contact pad **124** may include a second recessed side **126** coupled to a first terminal **144** of the second interconnect **112** at the first surface **104**.
(49) The second recessed side **126** of the second contact pad **124** may include a receding portion

that cuts inwards. In various aspects, the second contact pad **124** may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the second recessed side wall **114** and the second recessed side **126** may both have the same shape, e.g., a crescent shape as shown in FIG. **1**A and FIG. **1**B.

(50) In various aspects, the electronic assembly **100** may further include a subsequent second contact pad **128** arranged on the second surface **106** of the package substrate **102**. The subsequent second contact pad **128** may include a subsequent second recessed side **130** coupled to an opposing second terminal **146** of the second interconnect **112** at the second surface **106**.

(51) The subsequent second recessed side **130** of the subsequent second contact pad **128** may include a receding portion that cuts inwards. In various aspects, the subsequent second contact pad **128** may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the second recessed side wall **114** and the subsequent second recessed side **130** may both have the same shape, e.g., a crescent shape as shown in FIG. **1**A and FIG. **1**B.

(52) In one aspect, the second and the subsequent second recessed sides **126**, **130** may be aligned with the second recessed side wall **114** of the second interconnect **112**. In other words, the respective receding portion of the second recessed side **126** and the second recessed side wall **114** may coincide at the first surface **104** and the respective receding portion of the subsequent second recessed side **130** and the second recessed side wall **114** may coincide at the second surface **106**.

(53) In an aspect, the second recessed side **126** may be configured to face the first recessed side **118**. In a further aspect, the subsequent second recessed side **130** may be configured to face the subsequent first recessed side **122**.

(54) In various aspects, the electronic assembly **100** may include a plurality of first metal traces and/or planes **132** coupled to the first and the second contact pads **116**, **124**. In further aspects, the electronic assembly **100** may also include a plurality of second metal traces and/or planes **134** coupled to the subsequent first and the subsequent second contact pads **120**, **128**.

(55) FIG. **1**D shows PTH vertical interconnects of (i) a conventional design, and presently disclosed recessed design with (ii) a Recess Ø=10 mils and (iii) a Recess Ø=12 mils. The conventional design (i) may include a pitch of 21 mils; the recessed designs of (ii) and (iii) may each include a pitch of 16 mils.

(56) FIG. **1**E shows a simulation data of electrical performance comparison in insertion losses for the PTH vertical interconnects designs of FIG. **1**D. From the simulated results, it may be seen that the recessed designs of (ii) and (iii) may perform better than that of the conventional design (i). In particular, improvement in electrical performance is much higher after 30 GHz. The improved electrical (signal and/or power integrity) performance may be achieved through a tighter signal PTH to ground PTH coupling, i.e., improved signal current return path (for single-ended bus e.g., DDR memory interface) and a tighter differential signal PTHs coupling, e.g., a universal serial bus (USB) Interface, a peripheral component interconnect express (PCIe) interface or a high speed ethernet interface. Tighter power (Vcc) to ground (Vss) interconnect coupling reduces AC loop inductance for improved power supply delivery or power integrity.

(57) Thus, in one aspect, the first and the second interconnects **108**, **112** may be configured to facilitate a differential pair electrical signal, e.g., a universal serial bus Gen4 (USB4.0) Interface operating at ≥20 Gbps, a peripheral component interconnect express Gen5 (PCle5) interface operating at 32 Gbps or a serial-de-serializer (Serdes) ethernet interface operating at ≥112 Gbps.

(58) In another aspect, the first interconnect **108** may be configured to facilitate a single-ended electrical signal, e.g., a DDR memory interface at ≥5600 MT/s. In the same aspect, the second interconnect **112** may be configured to facilitate a reference voltage, e.g., a ground (Vss) reference voltage, for a reduced crosstalk coupling through a shorter current return path. In yet another aspect, the first interconnect **108** may be configured to facilitate a first reference voltage, e.g., a ground (Vss) reference voltage. In the same aspect, the second interconnect **112** may be configured

to facilitate a second reference voltage, e.g., a power supply (Vcc) reference voltage, for an improved power delivery through a shorter AC inductance loop.
(59) In various aspects, the electronic assembly **100** may include one or more devices (not shown) coupled to the first and the second interconnects **108**, **112** at the first surface **104**. The one or more devices may include semiconductor dies and/or packages, e.g., a central processing unit (CPU), a graphic processing unit (GPU), a memory controller, a field programmable gate array (FPGA), a neural network accelerator, a communication device, e.g., RFIC or a platform controller hub or chipset. In further aspects, the one or more devices may also be coupled to a system board (not shown) through the first and the second interconnects **108**, **112** through the second surface **106**.
(60) FIG. **2**A shows a perspective view of recessed vertical interconnects **208**, **212** in an electronic assembly **200** for improved electromagnetic field coupling according to another aspect of the present disclosure. FIG. **2**B shows a top view of the electronic assembly **200** according to the aspect as shown in FIG. **2**A. FIG. **2**C shows a cross-sectional view of the electronic assembly **200** according to the aspect as shown in FIG. **2**A.
(61) In various aspects, the electronic assembly **200** may include a package substrate **202** having a first surface **204** and an opposing second surface **206**. The first surface **204** may be a top surface of the package substrate **202**. The second surface **206** may be a bottom surface of the package substrate **202**. The first and second surfaces **204**, **206** may be substantially horizontal when viewed from a cross-sectional viewpoint. In one aspect, the package substrate **202** may include a bismaleimide-triazine (BT) epoxy layer, a polyimide layer, a glass substrate layer or a silicon substrate layer. In another aspect, the package substrate **202** may include two or more metal layers spaced apart by a dielectric layer.
(62) In various aspects, the package substrate **202** may include a first interconnect **208**. The first interconnect **208** may include a first side wall **210** having a length that may extend between the first and second surfaces **204**, **206** of the package substrate **202**. In one aspect, the first interconnect **208** may extend vertically across the thickness of the package substrate **202**. In other words, the first interconnect **202** may be perpendicular to the first and second surfaces **204**, **206** to form a first vertical interconnect **208**.
(63) In various aspects, the first side wall **210** may include a recessed portion that may extend between the first and second surfaces **204**, **206** of the package substrate **202** to form a first recessed side wall **210**. In the aspect shown in FIG. **2**A and FIG. **2**B, the first recessed side wall **210** of the first interconnect **208** may have a crescent-shaped cross section.
(64) In various aspects, the package substrate **202** may include a second interconnect **212**. The second interconnect **212** may include a second side wall **214** having a length that may extend between the first and second surfaces **204**, **206** of the package substrate **202**. In one aspect, the second interconnect **212** may extend vertically across the thickness of the package substrate **202**. In other words, the second interconnect **212** may be perpendicular to the first and second surfaces **204**, **206** to form a second vertical interconnect **212**.
(65) Different from the aspects shown in FIG. **1**A, FIG. **1**B, and FIG. **1**C, the second side wall **214** of the second interconnect **212** may not include a recessed portion.
(66) In various aspects, the first interconnect **208** may be arranged adjacent or close to, but not contacting, the second interconnect **212**. In one aspect, the first and the second interconnects **208**, **212** may be separated or kept apart by a dielectric or an insulation layer **240** arranged therebetween.
(67) In various aspects, the electronic assembly **200** may further include a first contact pad **216** arranged on the first surface **204** of the package substrate **202**. The first contact pad **216** may include a first recessed side **218** coupled to a first terminal of the first interconnect **208** at the first surface **204**.
(68) The first recessed side **218** of the first contact pad **216** may include a receding portion that cuts inwards. In various aspects, the first contact pad **216** may include a receding portion having a cross

section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the first recessed side wall **210** and the first recessed side **218** may both have the same shape, e.g., a crescent shape as shown in FIG. **2**A and FIG. **2**B.

(69) In various aspects, the electronic assembly **200** may further include a subsequent first contact pad **220** arranged on the second surface **206** of the package substrate **202**. The subsequent first contact pad **220** may include a subsequent first recessed side **222** coupled to an opposing second terminal of the first interconnect **208** at the second surface **206**.

(70) The subsequent first recessed side **222** of the subsequent first contact pad **220** may include a receding portion that cuts inwards. In various aspects, the subsequent first contact pad **220** may include a receding portion having a cross section in a crescent and/or gibbous shape. Other shapes may also be possible. Conveniently though not necessarily, the first recessed side wall **210** and the subsequent first recessed side **222** may both have the same shape, e.g., a crescent shape as shown in FIG. **2**A and FIG. **2**B.

(71) In one aspect, the first and the subsequent first recessed sides **218**, **222** may be aligned with the first recessed side wall **210** of the first interconnect **208**. In other words, the respective receding portion of the first recessed side **218** and the first recessed side wall **210** may coincide at the first surface **204** and the respective receding portion of the subsequent first recessed side **222** and the first recessed side wall **210** may coincide at the second surface **206**.

(72) In various aspects, the electronic assembly **200** may further include a second contact pad **224** arranged on the first surface **204** of the package substrate **202**. The second contact pad **224** may be coupled to a first terminal of the second interconnect **212** at the first surface **204**.

(73) In various aspects, the electronic assembly **200** may further include a subsequent second contact pad **228** arranged on the second surface **206** of the package substrate **202**. The subsequent second contact pad **228** may be coupled to an opposing second terminal of the second interconnect **212** at the second surface **206**.

(74) Different from the aspects shown in FIG. **1**A, FIG. **1**B, and FIG. **1**C, the second and the subsequent second contact pads **224**, **228** may not include a recessed side.

(75) In the aspect shown in FIG. **2**A and FIG. **2**B, the second interconnect **212** may be at least partially encircled by the first recessed side wall **210** of the first interconnect **208**. In other words, the second interconnect **212** may be arranged adjacent or close to, but not contacting, first recessed side wall **210** of the first interconnect **208**.

(76) Similar to the aspects shown in FIG. **1**A, FIG. **1**B, and FIG. **1**C, the electronic assembly **200** may include a plurality of first metal traces and/or planes (not shown) coupled to the first and the second contact pads **216**, **224**. In further aspects, the electronic assembly **200** may also include a plurality of second metal traces and/or planes (not shown) coupled to the subsequent first and the subsequent second contact pads **220**, **228**.

(77) In one aspect shown in FIG. **2**B, the first and the second interconnects **208**, **212** may be configured to facilitate a differential pair electrical signal, e.g., a universal serial bus Gen4 (USB4.0) interface operating at ≥20 Gbps, a peripheral component interconnect express Gen5 (PCle5) interface operating at 32 Gbps or a serial-de-serializer (Serdes) ethernet interface operating at ≥112 Gbps. For example, the first interconnect **208** may carry a P-signal while the second interconnect **212** may carry a complementary N-signal. When an electrical signal travels through the first and second interconnects **208**, **212**, such an arrangement may provide less signal distortions and/or coupling noises from adjacent signals, thereby improving the electrical integrity through a tighter coupling.

(78) In another aspect, the second interconnect **212** may be configured to facilitate a single-ended electrical signal, e.g., a DDR memory interface at ≥5600 MT/s. In the same aspect, the first interconnect **208** may be configured to facilitate a reference voltage, e.g., a ground (Vss) reference voltage, for a reduced crosstalk coupling through a shorter current return path. In yet another aspect, the first interconnect **208** may be configured to facilitate a first reference voltage, e.g., a

power supply (Vcc) reference voltage. In the same aspect, the second interconnect **212** may be configured to facilitate a second reference voltage, e.g., a ground (Vss) reference voltage, for an improved power delivery through a shorter AC inductance loop.

(79) In various aspects, the electronic assembly **200** may include one or more devices (not shown) coupled to the first and the second interconnects **208**, **212** at the first surface **204**. The one or more devices may include semiconductor dies and/or packages, e.g., a central processing unit (CPU), a graphic processing unit (GPU), a memory controller, a field programmable gate array (FPGA), a neural network accelerator, a communication device, e.g., RFIC or a platform controller hub or chipset. In further aspects, the one or more devices may also be coupled to a system board (not shown) through the first and the second interconnects **208**, **212** through the second surface **206**.

(80) FIGS. **3**A through **3**B show views directed to an exemplary simplified process flow for manufacturing the recessed vertical interconnects **308**, **312** according to an aspect that is generally similar to that shown in FIG. **1**A of the present disclosure. The order of assembly process operation may be interchangeable.

(81) FIG. **3**A shows a top view (top) and a cross-sectional view (bottom) of a first and a second interconnects **308**, **312** arranged adjacent each other in a package substrate or a printed circuit board **302**. The first interconnect **308** may include a first contact pad **316** and a subsequent first contact pad **320** coupled thereto as described in earlier paragraphs. Similarly, the second interconnect **312** may include a second contact pad **324** and a subsequent second contact pad **328** coupled thereto. A plurality of metal traces and/or planes **332**, **334** may also be coupled to the respective contact pads **316**, **320**, **324**, **328**. Conventional techniques may be employed, such as but not limited to, a photolithography, an etching, and a plating process.

(82) As may be seen in FIG. **3**A, the pitch between the first and second interconnects **308**, **312** may be smaller than that between interconnects of a conventional setup by arranging the presently disclosed contact pads as closely as possible. The first and second contact pads **316**, **324** may even be arranged to contact each other for further pitch reduction.

(83) FIG. **3**B shows a top view (top) and a cross-sectional view (bottom) of removal of an overlap contact pad section through, e.g., a mechanical or a laser drilling process. As shown in this figure, portions of the first, the subsequent first, the second, and the subsequent second contact pads **316**, **320**, **324**, **328** may be removed. A drilled through hole section (or recess opening) **336** may thus be formed in the package substrate or the printed circuit board **302**. An insulation or a dielectric layer **340**, e.g., an epoxy polymer resin layer, may be deposited within the recess opening **336** through, e.g., a plugging, a printing, a spraying, a coating or a dispensing process, followed by a subsequent solder resist deposition or lamination on top and bottom surfaces of the package substrate or the printed circuit board **302**.

(84) FIGS. **4**A through **4**G show views directed to an exemplary simplified process flow for manufacturing the recessed vertical interconnects **408**, **412** according to an aspect that is generally similar to that shown in FIG. **2**A of the present disclosure. The order of assembly process operation may be interchangeable.

(85) FIG. **4**A shows a cross-sectional view of a package substrate core layer **402** having a first and a second surfaces **404**, **406**. A first substrate core opening **403** may be formed by, e.g., a mechanical or a laser drilling process.

(86) FIG. **4**B shows formation of a first interconnect **408** in the first substrate core opening **403**. The formation may be performed through, e.g., an electroplating or a solder printing process. The left side of FIG. **4**B shows a cross-sectional view of the first interconnect **408** arranged in the package substrate core layer **402**. The right side of FIG. **4**B shows a perspective view of the first interconnect **408**. In one aspect, the first interconnect **408** may be a cylinder.

(87) FIG. **4**C shows formation of a second substrate core opening **405** in the package substrate core layer **402** through, e.g., a mechanical or a laser drilling process. During the drilling process, a portion of the first interconnect **408** may also be cut out to form a first recessed side wall **410** on

the first interconnect **408**. The left side of FIG. **4**C shows a cross-sectional view of the first interconnect **408** including the first recessed side wall **410** and second substrate core opening **405** in the package substrate core layer **402**. The right side of FIG. **4**C shows a perspective view of the first interconnect **408** including the first recessed side wall **410**. In one aspect, the first recessed side wall **410** may include a crescent-shaped cross section.

(88) FIG. **4**D shows a cross-sectional view of the package substrate core layer filled with a dielectric or an insulation layer **440**, e.g., an epoxy polymer resin layer, in the second substrate core opening **405**. The filling process may be carried out by e.g., a printing, a coating, a spraying, a dispensing, a plugging process.

(89) FIG. **4**E shows formation of a dielectric opening **436** through e.g., a mechanical or a laser drilling process. The dielectric opening **436** may be formed by drilling a portion of the dielectric layer **440**. The left side of FIG. **4**E shows a cross-sectional view of the first interconnect **408** including the first recessed side wall **410** and the dielectric opening **436** in the package substrate core layer **402**. The right side of FIG. **4**E shows a perspective view of the first interconnect **408** including the first recessed side wall **410**.

(90) FIG. **4**F shows formation of a second interconnect **412** in the dielectric opening **436** through e.g., an electroplating or a solder printing process. The left side of FIG. **4**F shows a cross-sectional view of the first and second interconnects **408**, **412** arranged in the package substrate core layer **402**. The right side of FIG. **4**F shows a perspective view of the first and second interconnects **408**, **412** arranged adjacent to each other. In one aspect, the second interconnect **412** may be a cylinder. The second interconnect **412** may be at least partially encircled by the first recessed side wall **410** of the first interconnect **408**. In other words, the second interconnect **412** may be arranged to face the first recessed side wall **410** of the first interconnect **408**. In one aspect, the first and the second surfaces **404**, **406** may be polished through e.g., a mechanical grinding process after the formation of the second interconnect **412**.

(91) FIG. **4**G shows formation of contact pads **416**, **420**, **424**, **428** on the package substrate core layer **402**. A first and a second contact pads **416**, **424** may be coupled to the first and the second interconnects **408**, **412**, respectively, on the first surface **404** through e.g., an electroplating, an etching process. A subsequent first and a subsequent second contact pads **420**, **428** may be coupled to the first and the second interconnects **408**, **412**, respectively, on the second surface **406** through e.g., an electroplating, an etching process. The left side of FIG. **4**G shows a cross-sectional view of the first and second interconnects **408**, **412** including the respective contact pads **416**, **420**, **424**, **428**. The right side of FIG. **4**G shows a perspective view of the first and second interconnects **408**, **412** including the respective contact pads **416**, **420**, **424**, **428**. In one aspect, a plurality of metal vias, traces and/or planes (not shown) may be coupled to the respective contact pads **416**, **420**, **424**, **428** through conventional techniques, such as but not limited to, a lamination, a drilling, a plating, a photolithography and an etching process.

(92) Aspects of the present disclosure may be implemented into a system using any suitable hardware and/or software. FIG. **5** schematically illustrates a computing device **500** that may include a semiconductor package as described herein, in accordance with some aspects. The computing device **500** may house a board such as a motherboard **502**. The motherboard **502** may include a number of components, including but not limited to a processor **504** and at least one communication chip **506**. The processor **504**, which may have an electronic assembly according to the present disclosure, may be physically and electrically coupled to the motherboard **502**. In some implementations, the at least one communication chip **506** may also be physically and electrically coupled to the motherboard **502**. In further implementations, the communication chip **506** may be part of the processor **504**.

(93) Depending on its applications, computing device **500** may include other components that may or may not be physically and electrically coupled to the motherboard **502**. These other components may include, but are not limited to, volatile memory (e.g. DRAM), non-volatile memory (e.g.

ROM), flash memory, a graphics processor, a digital signal processor, a cryptoprocessor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, a Geiger counter, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth).

(94) The communication chip **506** may enable wireless communications for the transfer of data to and from the computing device **500**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc. that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some aspects they might not. The communication chip **506** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 502.11 family), IEEE 502.16 standards (e.g., IEEE 502.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra-mobile broadband (UMB) project (also referred to as "3GPP2"), etc.). IEEE 502.16 compatible BWA networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 502.16 standards.

(95) The communication chip **506** may also operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UNITS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **506** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **506** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **506** may operate in accordance with other wireless protocols in other aspects.

(96) The computing device **500** may include a plurality of communication chips **506**. For instance, a first communication chip **506** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **506** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

(97) In various implementations, the computing device **500** may be a laptop, a netbook, a notebook, an ultrabook, a smartphone, a tablet, a personal digital assistant (PDA), an ultra-mobile PC, a mobile phone, a desktop computer, a server, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a digital camera, a portable music player, or a digital video recorder. In an aspect, the computing device **500** may be a mobile computing device. In further implementations, the computing device **500** may be any other electronic device that processes data.

(98) FIG. **6** shows a flow chart illustrating a method **600** of forming an electronic assembly according to an aspect of the present disclosure.

(99) As shown in FIG. **6**, at operation **602**, the method **600** of forming an electronic assembly may include providing a package substrate with a first surface and an opposing second surface.

(100) At operation **604**, the method may include arranging a first interconnect in the package substrate, the first interconnect extending between the first and the second surfaces, wherein the first interconnect may include a first recessed side wall.

(101) At operation **606**, the method may include arranging a second interconnect adjacent the first recessed side wall in the package substrate, the second interconnect extending between the first and the second surfaces.

(102) It will be understood that the above operations described above relating to FIG. **6** are not limited to this particular order. Any suitable, modified order of operations may be used.

EXAMPLES

(103) Example 1 may include an electronic assembly including a package substrate with a first surface and an opposing second surface; a first interconnect in the package substrate and may extend between the first and the second surfaces; and a second interconnect in the package substrate and may extend between the first and the second surfaces, wherein the first interconnect may include a first recessed side wall and the second interconnect may be arranged adjacent the first recessed side wall.

(104) Example 2 may include the electronic assembly of example 1 and/or any other example disclosed herein, further including a first contact pad including a first recessed side coupled to a first terminal of the first interconnect, and a subsequent first contact pad including a subsequent first recessed side coupled to an opposing second terminal of the first interconnect, wherein the first and the subsequent first recessed sides may be aligned with the first recessed side wall.

(105) Example 3 may include the electronic assembly of example 1 and/or any other example disclosed herein, wherein the second interconnect may be at least partially encircled by the first recessed side wall.

(106) Example 4 may include the electronic assembly of example 1 and/or any other example disclosed herein, wherein the second interconnect may further include a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces.

(107) Example 5 may include the electronic assembly of example 4 and/or any other example disclosed herein, wherein the first and the second recessed sides may face each other.

(108) Example 6 may include the electronic assembly of example 4 and/or any other example disclosed herein, further including a second contact pad including a second recessed side coupled to a first terminal of the second interconnect, and a subsequent second contact pad including a subsequent second recessed side coupled to an opposing second terminal of the second interconnect, wherein the second and the subsequent second recessed sides may be aligned with the second recessed side wall.

(109) Example 7 may include the electronic assembly of example 6 and/or any other example disclosed herein, further including a plurality of first metal traces or planes coupled to the first and the second contact pads, and a plurality of second metal traces or planes coupled to the subsequent first and the subsequent second contact pads.

(110) Example 8 may include the electronic assembly of example 1 and/or any other example disclosed herein, further including one or more devices coupled to the first and the second interconnects at the first surface.

(111) Example 9 may include the electronic assembly of example 1 and/or any other example disclosed herein, wherein the first and second interconnects may be configured to facilitate a differential pair electrical signal.

(112) Example 10 may include the electronic assembly of example 1 and/or any other example disclosed herein, wherein the first interconnect may be configured to facilitate a reference voltage and the second interconnect is configured to facilitate a single-ended electrical signal.

(113) Example 11 may include a computing device including a communication chip; and an electronic assembly coupled to the communication chip, the electronic assembly may include: a package substrate with a first surface and an opposing second surface; a first interconnect in the package substrate and may extend between the first and the second surfaces; and a second interconnect in the package substrate and may extend between the first and the second surfaces; wherein the first interconnect may include a first recessed side wall and the second interconnect may be arranged adjacent the first recessed side wall.

(114) Example 12 may include the computing device of example 11 and/or any other example disclosed herein, further including a first contact pad including a first recessed side coupled to a

first terminal of the first interconnect, and a subsequent first contact pad including a subsequent first recessed side coupled to an opposing second terminal of the first interconnect, wherein the first and the subsequent first recessed sides may be aligned with the first recessed side wall.
(115) Example 13 may include a computing device of example 11 and/or any other example disclosed herein, wherein the second interconnect may be at least partially encircled by the first recessed side wall.
(116) Example 14 may include the computing device of example 11 and/or any other example disclosed herein, wherein the second interconnect may further include a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces.
(117) Example 15 may include the computing device of example 14 and/or any other example disclosed herein, wherein the first and the second recessed sides may face each other.
(118) Example 16 may include the computing device of example 14 and/or any other example disclosed herein, further including a second contact pad including a second recessed side coupled to a first terminal of the second interconnect, and a subsequent second contact pad including a subsequent second recessed side coupled to an opposing second terminal of the second interconnect, wherein the second and the subsequent second recessed sides may be aligned with the second recessed side wall.
(119) Example 17 may include a method including providing a package substrate with a first surface and an opposing second surface; arranging a first interconnect in the package substrate, the first interconnect extending between the first and the second surfaces, wherein the first interconnect may include a first recessed side wall; and arranging a second interconnect adjacent the first recessed side wall in the package substrate, the second interconnect extending between the first and the second surfaces.
(120) Example 18 may include the method of example 17 and/or any other example disclosed herein, further including: coupling a first contact pad including a first recessed side to a first terminal of the first interconnect; coupling a subsequent first contact pad including a subsequent first recessed side to an opposing second terminal of the first interconnect; and aligning the first and the subsequent first recessed sides with the first recessed side wall.
(121) Example 19 may include the method of example 17 and/or any other example disclosed herein, further including at least partially encircling the second interconnect by the first recessed side wall.
(122) Example 20 may include the method of example 17 and/or any other example disclosed herein, further including: forming in the second interconnect a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces; coupling a second contact pad including a second recessed side to a first terminal of the second interconnect; coupling a subsequent second contact pad including a subsequent second recessed side to an opposing second terminal of the second interconnect; and aligning the second and the subsequent second recessed sides with the second recessed side wall.
(123) The term “comprising” shall be understood to have a broad meaning similar to the term “including” and will be understood to imply the inclusion of a stated integer or operation or group of integers or operations but not the exclusion of any other integer or operation or group of integers or operations. This definition also applies to variations on the term “comprising” such as “comprise” and “comprises”.
(124) The term “coupled” (or “connected”) used herein may be understood as electrically coupled or as mechanically coupled, e.g., attached or fixed or mounted, or just in contact without any fixation, and it will be understood that both direct coupling and indirect coupling (in other words, coupling without direct contact) may be provided.
(125) While the present disclosure has been particularly shown and described with reference to specific aspects, it should be understood by persons skilled in the art that various changes in form and detail may be made therein without departing from the scope of the present disclosure as

defined by the appended claims. The scope of the present disclosure is thus indicated by the appended claims and all changes which come within the meaning and range of equivalency of the claims are therefore intended to be embraced.

Claims

1. An electronic assembly comprising: a package substrate with a first surface and an opposing second surface; a first interconnect disposed in the package substrate and extending between the first and the second surfaces; a second interconnect disposed in the package substrate and extending between the first and the second surfaces; a first contact pad comprising a first recessed side coupled to a first terminal of the first interconnect; and a subsequent first contact pad comprising a subsequent first recessed side coupled to an opposing second terminal of the first interconnect, wherein the first interconnect comprises a first recessed side wall and the second interconnect is arranged adjacent the first recessed side wall, wherein the first and the subsequent first recessed sides are aligned with the first recessed side wall.
2. The electronic assembly of claim 1, wherein the second interconnect is at least partially encircled by the first recessed side wall.
3. The electronic assembly of claim 1, wherein the second interconnect further comprises a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces.
4. The electronic assembly of claim 3, wherein the first and the second recessed sides face each other.
5. The electronic assembly of claim 3, further comprising a second contact pad comprising a second recessed side coupled to a first terminal of the second interconnect, and a subsequent second contact pad comprising a subsequent second recessed side coupled to an opposing second terminal of the second interconnect, wherein the second and the subsequent second recessed sides are aligned with the second recessed side wall.
6. The electronic assembly of claim 5, further comprising a plurality of first metal traces or planes coupled to the first and the second contact pads, and a plurality of second metal traces or planes coupled to the subsequent first and the subsequent second contact pads.
7. The electronic assembly of claim 1, wherein the first and second interconnects are configured to facilitate a differential pair electrical signal.
8. The electronic assembly of claim 1, wherein the first interconnect is configured to facilitate a reference voltage and the second interconnect is configured to facilitate a single-ended electrical signal.
9. A computing device comprising: a communication chip; and an electronic assembly coupled to the communication chip, the electronic assembly comprising: a package substrate with a first surface and an opposing second surface; a first interconnect disposed in the package substrate and extending between the first and the second surfaces; a second interconnect disposed in the package substrate and extending between the first and the second surfaces; a first contact pad comprising a first recessed side coupled to a first terminal of the first interconnect; and a subsequent first contact pad comprising a subsequent first recessed side coupled to an opposing second terminal of the first interconnect, wherein the first interconnect comprises a first recessed side wall and the second interconnect is arranged adjacent the first recessed side wall, wherein the first and the subsequent first recessed sides are aligned with the first recessed side wall.
10. The computing device of claim 9, wherein the second interconnect is at least partially encircled by the first recessed side wall.
11. The computing device of claim 9, wherein the second interconnect further comprises a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces.

12. The computing device of claim 11, wherein the first and the second recessed sides face each other.

13. The computing device of claim 11, further comprising a second contact pad comprising a second recessed side coupled to a first terminal of the second interconnect, and a subsequent second contact pad comprising a subsequent second recessed side coupled to an opposing second terminal of the second interconnect, wherein the second and the subsequent second recessed sides are aligned with the second recessed side wall.

14. A method comprising: providing a package substrate with a first surface and an opposing second surface; arranging a first interconnect in the package substrate, the first interconnect extending between the first and the second surfaces, wherein the first interconnect comprises a first recessed side wall; arranging a second interconnect adjacent the first recessed side wall in the package substrate, the second interconnect extending between the first and the second surfaces; coupling a first contact pad comprising a first recessed side to a first terminal of the first interconnect; coupling a subsequent first contact pad comprising a subsequent first recessed side to an opposing second terminal of the first interconnect; and aligning the first and the subsequent first recessed sides with the first recessed side wall.

15. The method of claim 14, further comprising at least partially encircling the second interconnect by the first recessed side wall.

16. The method of claim 14, further comprising: forming in the second interconnect a second recessed side wall extending in parallel with the first recessed side wall between the first and the second surfaces; coupling a second contact pad comprising a second recessed side to a first terminal of the second interconnect; coupling a subsequent second contact pad comprising a subsequent second recessed side to an opposing second terminal of the second interconnect; and aligning the second and the subsequent second recessed sides with the second recessed side wall.
